01 / THE DEVELOPMENT

Native Phase-Word Reconstruction

Repeat the operation, recover the chart, retain the winding

The preceding time-comparison note exposed a modelling choice: an additive source phase and a linearly scaled Cayley coordinate obey different composition laws. This development uses actual repeated operators. It reconstructs their positive chart coordinates without assuming that doubling exposure doubles the Cayley parameter.

Let the native transports be $U = a\mathbf{1} + bR$ and $V = c\mathbf{1} + dJ$, with $a^2 + b^2 = c^2 + d^2 = 1$, $b, d > 0$, and anticommuting turn axes R, J. Their finite positive Cayley coordinates are q, p. For $m, n \geq 1$ form

$$H_{mn} = (V^n)^\dagger (U^m)^\dagger V^n U^m, \qquad (H_{mn} - \mathbf{1})^\dagger (H_{mn} - \mathbf{1}) = 4\chi_{mn}\mathbf{1}.$$

PW1–PW3: sufficient reconstruction protocol. Three normalized loop readings $\chi_{11}, \chi_{21}, \chi_{12}$ recover $u = b^2$ and $v = d^2$:

$$\boxed{u = 1 - \frac{\chi_{21}}{4\chi_{11}}, \qquad v = 1 - \frac{\chi_{12}}{4\chi_{11}}.}$$

Two oriented signs then select unique positive q, p. A fourth loop reading is predicted, without refitting:

$$\boxed{\chi_{22} = \frac{\chi_{21}\chi_{12}}{\chi_{11}}.}$$

Assumptions: $\chi_{11} > 0$, stable repeated transports, known oriented axes, and the inherited root-response normalization contract.

Every scalar integer-repeat loop remains blind to $q \leftrightarrow 1/q$ and $p \leftrightarrow 1/p$. The extra sign readouts resolve those branches while remaining insensitive to each transport's global phase. Consequently, a positive chart representative is recovered, but a physical unwrapped phase still needs a separate integer record.

Result type	Conditional algebraic inverse theorem and exact synthetic verification.
New in this packet	Three-reading amplitude inversion; oriented branch selection; repeated-word predictions; explicit distinction between phase coupling and linear coordinate gain.
Inherited	Native cut arithmetic, anticommuting cell, Cayley composition, root-response normalization and phase-continuation machinery.
Verification	13 polynomial certificates and replays; 952 checks; 13 negative controls; 25 cells and 400 integer-word cases.
Physical status	No measured coupling, value of G, empirical gravity result or completed quantum-gravity theory is claimed.

02 / NATIVE OBJECTS

The word law survives chart poles

Use the inherited exact cut scalar $\iota^2 = -1$ and finite path-composition coordinates

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad K = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad J = \iota K, \quad Z = -RK = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Then $R^2 = J^2 = -\mathbf{1}$, $RJ = -JR$, while K, ιR and Z are self-adjoint. These matrices realize the native algebra; they do not assume a primitive physical Hilbert space or physical time.

$$U(q) = (\mathbf{1} - qR)^{-1}(\mathbf{1} + qR) = a\mathbf{1} + bR, \quad a = \frac{1-q^2}{1+q^2}, \quad b = \frac{2q}{1+q^2},$$

and similarly $V(p) = c\mathbf{1} + dJ$. For finite $q, p > 0$, $b, d > 0$.

PW1. Repeated-word factorization. Write $U^m = a_m\mathbf{1} + b_m R$ and $V^n = c_n\mathbf{1} + d_n J$. Multiplication and $a^2 + b^2 = 1$ give

$$\begin{pmatrix} a_{m+1} \\ b_{m+1} \end{pmatrix} = \begin{pmatrix} a & -b \\ b & a \end{pmatrix} \begin{pmatrix} a_m \\ b_m \end{pmatrix}, \qquad b_{m+1} = 2ab_m - b_{m-1}, \quad b_0 = 0, \; b_1 = b.$$

The identical recurrence holds for d_n with coefficient c. For generic unit pairs, direct native multiplication gives

$$H = (1 - 2b^2d^2)\mathbf{1} + 2abd^2R - 2b^2cdJ - 2abcdRJ.$$

Therefore $(H - \mathbf{1})^\dagger(H - \mathbf{1}) = 4b^2d^2\mathbf{1}$. Apply this identity to (U^m, V^n):

$$\boxed{\chi_{mn} = b_m^2 d_n^2.}$$

The executable supplies four entry certificates for the expansion and four for the defect identity. The all-integer extension follows by the written recurrence and closure of unit pairs.

How the invariant is read

For the inherited unit-weight four-cycle, let S carry edges $U^m, V^n, (U^m)^\dagger, (V^n)^\dagger$ and $L = g(2\mathbf{1} - S - S^\dagger)$. Eliminating the six interior coordinates gives the root blocks

$$K_{\mathrm{bare}} = 2g\mathbf{1}_2, \qquad K_{\mathrm{eff}} = g\chi_{mn}\mathbf{1}_2, \qquad \chi_{mn} = \frac{2\,\mathrm{tr}\,K_{\mathrm{eff}}}{\mathrm{tr}\,K_{\mathrm{bare}}}.$$

The positive gain may differ between frames, but must be shared within each bare/effective pair. The interior block is invertible even at $\chi_{mn} = 0$; the full L need not be. This is a stipulated stationary-response interface, not a claim that channel tomography alone supplies this response.

03 / THE INVERSE THEOREM

Three readings determine amplitudes

PW2. Amplitude admission and reconstruction. Within the positive finite base chart, the three readings determine $u = b^2$, $v = d^2$ by

$$x := \chi_{11}, \quad y := \chi_{21}, \quad z := \chi_{12}, \qquad u = 1 - \frac{y}{4x}, \quad v = 1 - \frac{z}{4x}.$$

They arise from at least one positive-chart cell if and only if

$$\boxed{x > 0, \qquad 0 < u, v \le 1, \qquad x = uv.}$$

These conditions also enforce $0 \le y, z \le 1$. The repeated readings y, z are allowed to vanish.

Proof. The recurrence yields $b_2 = 2ab$, $d_2 = 2cd$, so

$$x = uv, \qquad y = 4(1-u)uv, \qquad z = 4(1-v)uv.$$

Division by $4x$ proves necessity and the formulas. Conversely, choose

$$b = \sqrt{u}, \quad a = \pm\sqrt{1-u}, \qquad d = \sqrt{v}, \quad c = \pm\sqrt{1-v}.$$

The signs coincide when the corresponding square root vanishes. Then $q = (1-a)/b$ and $p = (1-c)/d$ are finite and positive, satisfy the unit-pair relations, and reproduce all three readings. No hidden generating parameter is supplied to the decoder. □

All integer-word predictions follow

Set $P_0(a) = 1$, $P_1(a) = 2a$ and $P_{j+1}(a) = 2aP_j(a) - P_{j-1}(a)$. The native recurrence gives

$$b_m = bP_{m-1}(a), \qquad \chi_{mn} = xP_{m-1}(a)^2 P_{n-1}(c)^2.$$

In particular,

$$\boxed{\chi_{22} = yz/x, \quad \chi_{31} = x(3-4u)^2, \quad \chi_{13} = x(3-4v)^2.}$$

More generally, $\chi_{mn}\chi_{11} = \chi_{m1}\chi_{1n}$. The word-response array has rank one when its nonzero base entry is included. This separability concerns this fixed two-axis cell, not arbitrary noncommuting transport networks.

Exact theorem versus executable domain. The theorem uses positive square roots in the inherited native real completion. The packaged finite engine executes exact rational cases only. Its root adapter checks for perfect rational squares and refuses a floating-point fallback. No noisy-data estimator or general real-completion implementation is asserted here.

04 / THE MISSING OBSERVATION

Two oriented signs select the branch

Define the native comparison moments

$$w_q = \frac{1}{2}\operatorname{tr}\big(ZU^\dagger KU\big)\,, \qquad w_p = \frac{1}{2}\operatorname{tr}\big(ZV^\dagger(\iota R)V\big)\,.$$

Direct multiplication, certified in the unit-pair quotient, gives

$$U^\dagger KU = (a^2-b^2)K + 2abZ, \qquad V^\dagger(\iota R)V = (c^2-d^2)(\iota R) + 2cdZ,$$

and hence $w_q = 2ab$, $w_p = 2cd$.

PW3. Oriented positive-chart reconstruction. Since $b, d > 0$, the signs $\epsilon_q = \operatorname{sgn}(w_q)$ and $\epsilon_p = \operatorname{sgn}(w_p)$ give

$$a = \epsilon_q\sqrt{1-u}, \quad b = \sqrt{u}, \qquad c = \epsilon_p\sqrt{1-v}, \quad d = \sqrt{v},$$

$$\boxed{q = \frac{1-a}{b}, \qquad p = \frac{1-c}{d}.}$$

Admission requires $\epsilon_q = 0$ exactly when $u = 1$, and $\epsilon_p = 0$ exactly when $v = 1$. Otherwise each sign is $+1$ or -1. The continuous moment magnitudes are not inputs to the decoder. If available, they supply the extra checks $w_q^2 = 4u(1-u)$ and $w_p^2 = 4v(1-v)$.

Why integer repetitions alone cannot choose

The reciprocal transformation $q \mapsto 1/q$ sends $(a, b) \mapsto (-a, b)$. The recurrence proves by induction

$$b_m \mapsto (-1)^{m+1} b_m.$$

Thus every χ_{mn} is unchanged. The same argument applies independently to p. Conversely PW2 determines b^2, d^2, and for positive q, p each allows only the two reciprocal choices. The entire integer-word catalogue therefore has exactly these four generic branches, with coincident branches at $q = 1$ or $p = 1$.

Positive branch	ϵ_q	ϵ_p	Scalar integer-word catalogue
$(1/2, 1/3)$	$+1$	$+1$	Identical for all four rows
$(2, 1/3)$	-1	$+1$	Identical
$(1/2, 3)$	$+1$	-1	Identical
$(2, 3)$	-1	-1	Identical

The oriented moments survive independent global phases on U, V. They also survive a joint unitary frame change when the transports and $K, \iota R, Z$ are conjugated together. An unknown positive detector gain preserves a sign; an unknown offset or comparator orientation does not. These are additional observable interfaces, not information inferred from the scalar loop alone.

05 / AN EXACT WORKED CASE

Recover first, predict the next word

Generate the synthetic native cell $q = 1/2$, $p = 1/3$:

$$(a, b) = (3/5, 4/5), \qquad (c, d) = (4/5, 3/5).$$

The decoder receives only the following normalized comparison outputs. Generating parameters and moment magnitudes are withheld from it.

Transport pair	Reading	Purpose
(U, V)	$\chi_{11} = 144/625$	Base response
(U^2, V)	$\chi_{21} = 5184/15625$	First-axis repetition
(U, V^2)	$\chi_{12} = 9216/15625$	Second-axis repetition
First oriented channel	$\epsilon_q = +1$	First reciprocal branch
Second oriented channel	$\epsilon_p = +1$	Second reciprocal branch

PW2 gives

$$u = 1 - \frac{5184/15625}{4(144/625)} = \frac{16}{25}, \qquad v = 1 - \frac{9216/15625}{4(144/625)} = \frac{9}{25}.$$

The signs select $a = 3/5$, $c = 4/5$, and PW3 returns

$$q = \frac{1 - 3/5}{4/5} = \frac{1}{2}, \qquad p = \frac{1 - 4/5}{3/5} = \frac{1}{3}.$$

The unused fourth response must be

$$\boxed{\chi_{22} = \frac{(5184/15625)(9216/15625)}{144/625} = \frac{331776}{390625}.}$$

An independently evaluated native Schur response at (U^2, V^2) gives exactly this fraction. The further held-out words give

$$\chi_{31} = \frac{17424}{390625}, \qquad \chi_{13} = \frac{219024}{390625}.$$

Unknown gain was actually varied

The three fitted normalized responses use gains $2, 3, 5$, and the held-out response uses gain 7. Each response is calculated from the full native eight-coordinate operator before eliminating the interior. Thus normalization is exercised, rather than assuming that all measurements had an identical gain.

This is a synthetic identifiability result. The agreement verifies the calculation under the declared algebra and readout contract. It is not an experimental observation of a gravitational interaction. A real implementation must supply stable repeated transports, paired root responses and oriented comparators.

06 / SOURCE ACTION TO PHASE

Additivity fixes the local coordinate law

Let x be an independently specified additive source-action comparison. It need not be a primitive time variable. Suppose a same-axis continuation satisfies

$$U(x+y) = U(x)U(y), \qquad U(0) = \mathbf{1},$$

is differentiable at zero, and has a finite Cayley chart $q(x)$ near zero with $q(0) = 0$. The inherited Cayley product law then requires

$$q(x+y) = \frac{q(x) + q(y)}{1 - q(x)q(y)}$$

where the denominator is nonzero. Put $g = q'(0)$. Differentiating in y at zero yields

$$\boxed{q'(x) = g\big(1 + q(x)^2\big).}$$

The native first-order jet certificate, with central e and $e^2 = 0$, is

$$\big[q + g(1+q^2)e\big](1 - qge) = q + ge \pmod{e^2}.$$

This verifies the algebraic derivative step. Differentiability, integration and a continued real phase are stated analytic assumptions, not extra conclusions of the finite polynomial certificate.

Define phase through the native continuation

Using the inherited real-completion integration, define

$$\Phi(q) = 2\int_0^q \frac{du}{1+u^2}.$$

The chain rule gives $d\Phi(q(x))/dx = 2g$. On the connected chart containing zero,

$$\boxed{\Phi(q(x)) = \kappa x, \qquad \kappa = 2g.}$$

Equivalently write $q(x) = \tan_\Sigma(\kappa x/2)$, where $\tan_\Sigma$ is the inverse of $q \mapsto \Phi(q)/2 = \int_0^q du/(1+u^2)$, with the chosen branch. No numerical trigonometric backend is used in the verifier. Full operators and a retained continuation record are needed across chart poles.

A constant phase coupling is a different adapter

The earlier proposal $q = \alpha x$ is a linear Cayley-input dictionary. If the same x is additive under operator composition, a constant nonzero α cannot satisfy that law globally. Instead

$$\alpha_{\mathrm{eff}}(x) = \frac{q(x)}{x} = \frac{\tan_\Sigma(\kappa x/2)}{x}, \qquad \lim_{x\to 0}\alpha_{\mathrm{eff}}(x) = \frac{\kappa}{2}.$$

For example, $q = 1/2$ repeated twice gives $q_{\mathrm{join}} = 4/3$, not 1. The earlier four-reading fixed-scaling theorem remains valid under its own linear-input control assumption; it must not be relabelled as an additive-phase experiment. The present repetition protocol avoids that assumption.

07 / PHYSICAL ADAPTER AND MEMORY

Keep the unresolved integer visible

The present observables are invariant under $U \mapsto -U$ and $V \mapsto -V$. Indeed,

$$U(-1/q) = -U(q), \qquad V(-1/p) = -V(p),$$

and those central signs cancel in every commutator word and oriented channel moment. The positive restriction $q, p > 0$ chooses representatives; it does not measure those signs.

With $\Phi(q) = 2\int_0^q du/(1+u^2)$, define the half-turn constant by $\pi := \lim_{q\to+\infty} \Phi(q)$. A full unit-pair turn has period 2π; a channel is insensitive to its central half-turn. Thus the unwrapped transport angles remain

$$\boxed{\theta_Q = \Phi(q) + \pi m_Q, \qquad \theta_G = \Phi(p) + \pi m_G, \qquad m_Q, m_G \in \mathbb{Z}.}$$

Only a separately justified branch bound or a continuous, oriented unwrapping record permits choosing the integers. A coherent global-phase reference can distinguish the central sign, but still does not erase ordinary full-turn winding. Finitely many dyadic subdivisions also leave aliases unless supplemented by a bound or continuation information.

Optional time-comparison dictionary

For a specified photon, use $E = \hbar\omega$ [1]. Put $\Theta = 1/\omega$ and, for a calibrated vacuum-flight path, $\Delta = \ell/c$ [2]. These established relations supply comparisons; they do not establish a gravitational transport coupling.

A *proposed phase version* of the preceding dictionary would be

$$\theta_Q = \kappa_Q \frac{E\ell}{\hbar c} = \kappa_Q \frac{\Delta}{\Theta}, \qquad \theta_G = \kappa_G \frac{GE}{c^4\ell} = \frac{\zeta_{\mathrm{ph}}}{\Delta\Theta}, \qquad \zeta_{\mathrm{ph}} := \kappa_G \frac{\hbar G}{c^5}.$$

It would imply

$$\boxed{\kappa_Q = [\Phi(q) + \pi m_Q]\frac{\Theta}{\Delta}, \qquad \zeta_{\mathrm{ph}} = [\Phi(p) + \pi m_G]\Delta\Theta.}$$

These are conditional calibration formulas. In particular, $GE/(c^4\ell)$ has not been shown to be an additive action for a realized gravitational operation. Repeating V is a word operation, not automatically a longer path or a change of ℓ. An experimental source-to-transport map remains to be supplied.

Retained record	What the current reconstruction determines
Positive chart	q, p from loop amplitudes and oriented signs.
Winding/central phase	m_Q, m_G remain unmeasured by this protocol.
Clock conversion	Θ/Δ is unchanged by a common clock gain; $\Delta\Theta$ scales by its square.
Gravity calibration	Even with a branch record, only the proposed product $\kappa_G\hbar G/c^5$ is obtained, not G separately.

08 / BOUNDARIES AND REJECTION TESTS

The selected cell has a parity signature

At the inherited positive two-cut-compatible cell $q = p = 1$, the transports are $U = R$, $V = J$. They anticommute, and therefore

$$H_{mn} = (-1)^{mn}\mathbf{1}, \qquad \chi_{mn} = \begin{cases} 1, & m, n \text{ both odd}, \\ 0, & \text{otherwise}. \end{cases}$$

This is an exact consequence of the earlier quarter-turn algebra, not a new derivation of fundamental quantisation.

χ_{mn}	$n=1$	$n=2$	$n=3$	$n=4$
$m=1$	1	0	1	0
$m=2$	0	0	0	0
$m=3$	1	0	1	0
$m=4$	0	0	0	0

Here $(\chi_{11}, \chi_{21}, \chi_{12}) = (1, 0, 0)$ and the two oriented signs are zero. The decoder returns $q = p = 1$. Doubling gives $U^2 = -\mathbf{1}$: the fractional Cayley coordinate has a pole while the operator stays finite. Repeated words continue through this point without dividing by $1 - q^2$.

An operational protocol

1. Fix the two oriented axes, comparator conventions and zero offsets. Verify that each repeated word uses the same transport.
2. Obtain the three normalized root responses for $(U, V), (U^2, V), (U, V^2)$. Each normalized response requires its paired bare/effective readings.
3. Compute u, v and apply the admission conditions. Obtain the two oriented signs and reconstruct positive q, p.
4. Predict (U^2, V^2), then read it. Check the further triple-word predictions when they address a remaining model uncertainty.
5. If physical coupling is sought, retain the source-action definition, clock conversion and phase-unwrapping integers separately.

This is a sufficient protocol, not a claim of the fewest physically possible measurements.

What a disagreement means

Failure of $\chi_{22}\chi_{11} = \chi_{21}\chi_{12}$ rejects at least one joint assumption: fixed transports, two fixed anticommuting axes, or the readout normalization. It does not identify which assumption failed. Passing a finite set of checks is also insufficient to establish the full physical model.

For real noisy readings, exact equality tests must become uncertainty regions. In particular

$$\delta u \simeq -\frac{\delta\chi_{21}}{4\chi_{11}} + \frac{\chi_{21}\,\delta\chi_{11}}{4\chi_{11}^2}.$$

Small base response amplifies error; a sign near zero requires a resolved zero convention. The present verifier uses exact synthetic inputs and does not claim a noise-certified instrument protocol.

09 / PROOF AND SOURCE AUDIT

What was checked, and what remains

All mathematical execution uses the unchanged pinned native cut engine and native rewrite compiler. Python 3.12 performs file integrity and process orchestration only. There is no NumPy, SymPy, floating-point trigonometry or second mathematical engine.

Executable evidence	Coverage
13 polynomial certificates	Four commutator entries, four defect entries, two oriented moments, double/triple amplitudes and one first-order continuation jet.
13 serialized replays	Certificates are serialized and replayed against the declared quotient; both presentations pass confluence and dagger gates.
952 exact checks	Includes certificate checks, all data-only reconstructions, held-out predictions, integer words, frame/phase invariance and balanced-cell parity.
25 synthetic cells	$q, p \in \{1/3, 1/2, 1, 2, 3\}$, including reciprocal branches and the chart-pole base point.
400 word cases	$1 \le m, n \le 4$ for each cell. Finite controls support, but do not replace, the all-integer induction proof.
13 negative controls	Invalid data/signs, zero base, irrational root refusal, reciprocal and central-sign ambiguity, naive coordinate doubling, chart pole, offset error, drift and incompatible linear calibration.

Dependency and duplication boundary

The unchanged parent packet is *Clock-Relation Reconstruction* (v0.5), with the scale compass, physics cubits, quantisation packet and native lattice nested beneath it. The native library's GP1 translator already treats calibrated phase-referenced inversion; GP2 already records global-phase blindness. GE2 already develops native phase refinement and its retained ledger. Those results are reused, not claimed as discoveries here.

The advance here is a specific inverse observation protocol for repeated two-axis words, its complete positive-chart ambiguity, and explicit phase-versus-coordinate calibration conditions. General unit-circle recurrences and Cayley composition are established algebraic identities; no broad mathematical priority claim is made.

Reproduce

Extract the source archive and run `python3.12 verify.py` from its packet directory. This verifies all nested manifests and source pins, checks the unchanged engine hashes, reruns the native verifier and compares the complete result JSON. The TeX source, protocol JSON, certificate records and parent packets are included. Analytic continuation is a written conditional proof, not a proof-assistant certificate.

The inherited canonical source anchor is Recognition-Kernel-Framework commit:

`e8e3089745bf7dc0532d01c62a5f660f5658c6f0`.

The native foundation is under `operator_foundation/`. The inherited pin files retain exact paths and hashes, including separately pinned sources. No repository files were changed by this development.

Physical references and next target

[1] NIST, *Kilogram: Mass and Planck's Constant*: photon energy–frequency relation.

[2] BIPM, *SI base unit: metre*: vacuum light-speed relation. Accessed 14 October 2026. These references support only the stated physical inputs.

The next unresolved research target is a bounded, oriented phase-unwrapping protocol tied to a realizable source control, followed by a specified gravitational coupling test. Neither a missing integer nor an unknown coupling is fixed by naming it an invariant; each needs an additional observable constraint.